

EXPERIMENT 4 – CPU Scheduling Algorithms

Aim: To implement FCFS, SJF, Priority Scheduling, and Round Robin algorithms.

1. First Come First Serve (FCFS)

C Program:

```
#include<stdio.h>

int main()
{
    int n, i;
    int bt[20], wt[20], tat[20];
    float avg_wt=0, avg_tat=0;

    printf("Enter Number of Processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
    }

    wt[0]=0;

    for(i=1;i<n;i++)
        wt[i]=wt[i-1]+bt[i-1];

    for(i=0;i<n;i++)
    {
        tat[i]=wt[i]+bt[i];
        avg_wt+=wt[i];
        avg_tat+=tat[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);

    printf("\nAverage Waiting Time = %.2f",avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n",avg_tat/n);

    return 0;
}
```

Output (Input: 4 processes, Burst Times: 6 4 12 5):

Process	BT	WT	TAT
P1	6	0	6
P2	4	6	10
P3	12	10	22
P4	5	22	27

Average Waiting Time = 9.50
Average Turnaround Time = 16.25

2. Shortest Job First (SJF)

C Program:

```
#include<stdio.h>

int main()
{
    int n, i, j, temp;
    int bt[20], wt[20], tat[20];
    float avg_wt=0, avg_tat=0;

    printf("Enter Number of Processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
    }
}
```

```

}

for(i=0;i<n-1;i++)
    for(j=i+1;j<n;j++)
        if(bt[i] > bt[j])
        {
            temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
        }

wt[0] = 0;

for(i=1;i<n;i++)
    wt[i] = wt[i-1] + bt[i-1];

printf("\nProcess\tBT\tWT\tTAT\n");

for(i=0;i<n;i++)
{
    tat[i] = wt[i] + bt[i];
    avg_wt += wt[i];
    avg_tat += tat[i];
    printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
}

printf("\nAverage Waiting Time = %.2f", avg_wt/n);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);

return 0;
}

```

Output (Input: 4 processes, Burst Times: 6 4 12 5):

Process	BT	WT	TAT
P1	4	0	4
P2	5	4	9
P3	6	9	15
P4	12	15	27

Average Waiting Time = 7.00
Average Turnaround Time = 13.75

3. Priority Scheduling

C Program:

```

#include<stdio.h>

int main()
{
    int n, i, j, temp;
    int bt[20], pr[20], wt[20], tat[20];
    float avg_wt=0, avg_tat=0;

    printf("Enter Number of Processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("\nEnter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
        printf("Enter Priority for P%d: ",i+1);
        scanf("%d",&pr[i]);
    }

    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(pr[i] > pr[j])
            {
                temp=pr[i]; pr[i]=pr[j]; pr[j]=temp;
                temp=bt[i]; bt[i]=bt[j]; bt[j]=temp;
            }

    wt[0] = 0;

    for(i=1;i<n;i++)
        wt[i] = wt[i-1] + bt[i-1];

    printf("\nProcess\tPriority\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)

```

```

{
    tat[i] = wt[i] + bt[i];
    avg_wt += wt[i];
    avg_tat += tat[i];
    printf("P%d\t%d\t%d\t%d\t%d\n", i+1, pr[i], bt[i], wt[i], tat[i]);
}

avg_wt /= n; avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

return 0;
}

```

Output (Input: 3 processes — P1: BT=10 Pri=3, P2: BT=1 Pri=1, P3: BT=4 Pri=4):

Process	Priority	BT	WT	TAT
P1	1	1	0	1
P2	3	10	1	11
P3	4	4	11	15

Average Waiting Time = 4.00
Average Turnaround Time = 9.00

4. Round Robin Scheduling

C Program:

```

#include<stdio.h>

int main()
{
    int n, tq, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int time=0, done;
    float avg_wt=0, avg_tat=0;

    for(i=0;i<20;i++) wt[i]=0;

    printf("Enter Number of Processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
        rem_bt[i]=bt[i];
    }

    printf("Enter Time Quantum: ");
    scanf("%d",&tq);

    do {
        done=1;
        for(i=0;i<n;i++)
        {
            if(rem_bt[i]>0)
            {
                done=0;
                if(rem_bt[i]>tq) { time+=tq; rem_bt[i]-=tq; }
                else { time+=rem_bt[i]; wt[i]=time-bt[i]; rem_bt[i]=0; }
            }
        }
    } while(!done);

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
    {
        tat[i]=bt[i]+wt[i];
        avg_wt+=wt[i];
        avg_tat+=tat[i];
        printf("P%d\t%d\t%d\t%d\n",i+1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);

    return 0;
}

```

}

Output (Input: 3 processes BT=24,3,3 Time Quantum=4):

Process	BT	WT	TAT
P1	24	6	30
P2	3	4	7
P3	3	7	10

Average Waiting Time = 5.67

Average Turnaround Time = 15.67

Result: CPU Scheduling Algorithms (FCFS, SJF, Priority, Round Robin) were implemented successfully.
